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Title: V84 - The NBA's Tactical Evolution

1. Objectives

Playstyle, player roles, and scoring tactics have undergone major shifts in the NBA, mainly due to the rise of the 3-point shot. Age dynamics, positional demands, and the development of individual players have all been redefined as a result of this change. Therefore, the first goal of this project is to determine how the game evolved over time, and the second is to determine how players have adapted to stay up with the game's ever-evolving pace. To make it more interesting, Kobe Bryant's career performance serves as an example of how data storytelling may uncover deeper human stories hidden behind the numbers.

2. Novelty

This project reflects my curiosity to explore visual storytelling beyond standard chart types taught in the module. I incorporated new and creative visualizations such as using basketball symbols to represent number of 3-Point Makes and 3-Point Attempts, comparing the performances between the two 3-Point legends (Stephen Curry and James Harden). Sunburst Chart and Lollipop Chart are created to communicate multi-layered insights effectively. I am particularly proud of the Sunburst Chart, which captures Kobe Bryant's career evolution across three distinct phases: Rookie (Age: 18–24), Prime (Age: 25–34), and Veteran (Age: 35–37). The Sunburst layout in Figure 1 elegantly displays hierarchical data across three levels:

Table 1. Sunburst layout representing performance metrics across Kobe Bryant's career phases.

Level	Attributes/ Representations
Level 1	Career phase (Rookie, Prime, Veteran)
Level 2	Performance metrics including Field Goal Percentage (FG%), Free Throw Percentage (FT%), True Shooting Percentage (TS%)
Level 3	Comparison between Kobe and the Lakers team average

This design allows for intuitive exploration where we can zoom into “Prime (25–34)” to see Kobe's efficiency relative to the Lakers' team averages as shown in Figure 2. To enhance readability, I used a continuous colour palette where darker blue indicates higher shooting percentage and lighter blue represents lower ones, helping users quickly interpret performance differences.

Kobe Bryant's Career Phase Performance

Kobe Bryant vs Los Angeles Lakers

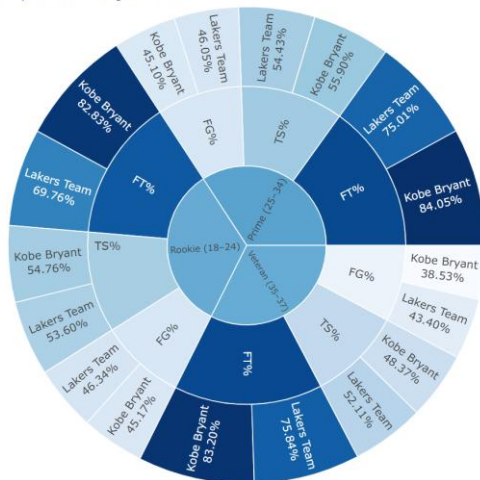


Figure 1

Kobe Bryant's Career Phase Performance

Kobe Bryant vs Los Angeles Lakers

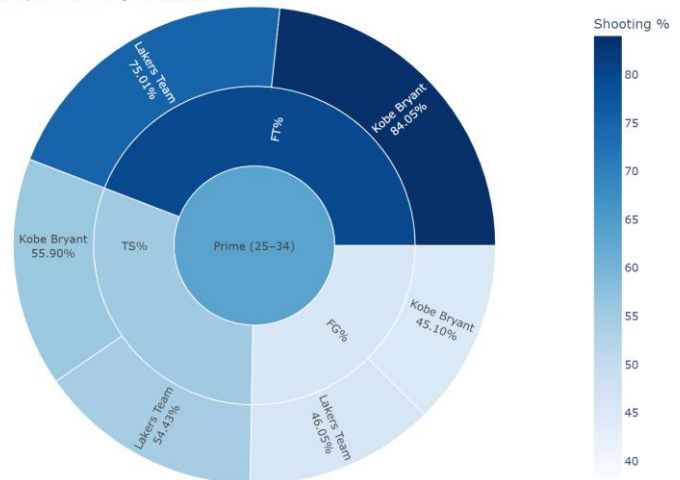


Figure 2

Besides that, I added a new attribute called True Shooting Percentage to calculate the shooting efficiency of a player using the following formula:

$$TS\% = \frac{Total\ Points}{2 \times (Total\ FGA + 0.44 \times Total\ FTA)}$$

where FGA represents Field Goal Attempt and FTA represents Free Throw Attempt. This metric provides a more accurate measure of overall shooting efficiency by considering both field goals and free throws, enriching the depth of performance analysis.

3. Technical Challenges and Innovation

During data cleaning and engineering, I performed column renaming, datatype conversion, and data aggregation such as grouping data into different age groups and phases. Most of the operations are done using the Pandas library. NumPy is used to perform numerical operations. Functions such as `groupby()`, `agg(mean/sum/count)`, `concat()` etc. are used to prepare meaningful data that fits the visualisation tasks. Plotly and Matplotlib libraries are utilised to create interesting visuals including multi-line graph, stacked bar chart, scatter plot with 4 variables, boxplot, sunburst chart and lollipop chart. The basketball symbols diagram is created on Canva.

Each visualization was carefully designed based on principles of effective visual communication. I incorporated redundant encoding, such as adding trend lines over bar plots, to reinforce key insights. The principles of similarity and proximity were applied to align colours and visually group related elements such as using the same colour for axis, annotation and visual marks that are related. The principle of enclosure was used to highlight critical sections with semi-transparent red boxes. To emphasize focal insights, I employed the principle of salience using arrows, speech bubbles/ text boxes, as well as annotations in the presentation video. Additionally, I ensured strong figure-ground contrast by using muted grey tones for non-essential details and vivid hues for key data points. To make the charts look simple and clean, I removed unnecessary axes and grid lines, keeping the viewer's attention centred on the main narrative conveyed by the visuals.

A major technical challenge I overcame was web scraping, where I extracted player data from Basketball Reference using the BeautifulSoup Python library. It was my first time implementing data scraping, which required careful handling of HTML structures and data cleaning. Despite the challenge, this step enabled me to access rich, authentic datasets that enhanced the storytelling power of the final visualizations.

Another technical challenge is that the initial plots were heavily cluttered due to the massive volume of NBA game data spanning from the 1983 to 2023 seasons. To tackle this, I narrowed the project's scope and applied indexing and conditional selection techniques using NumPy to filter the data effectively. By studying best practices in data visualization and continuously refining my designs, I managed to create clear, engaging, and insightful visuals that not only addressed the problem statement but also brought the story of the NBA's evolution to life through data-driven storytelling.

4. References

- [1] Swar, S. (n.d.). NBA_API. GitHub. https://github.com/swar/nba_api/tree/master/docs/nba_api
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- [3] Mathur, A. (2020, June 30). Looking back at Kobe Bryant's career after his Achilles tear. ClutchPoints. <https://clutchpoints.com/nba/los-angeles-lakers/looking-back-at-kobe-bryant-career-after-his-achilles-tear>